Refining Local Descriptors by Embedding Semantic Information for Visual Categorization*

Yingbin Zheng, Renzhong Wei, Hong Lu†, Xiangyang Xue
Shanghai Key Lab of Intel. Info. Processing
School of Computer Science
Fudan University, Shanghai, China
{ybzh, 0456078, honglu, xyxue}@fudan.edu.cn

ABSTRACT

Local descriptor extraction and vector quantization are the important components of widely-used Bag-of-Features (BoF) model for visual categorization. This paper proposes a simple and efficient approach to refine the local descriptors for vector quantization by embedding semantic information. The original local descriptors are integrated by a sequence of category-independent and category-dependent basis. Particularly, the category-dependent basis is learned by minimizing the joint loss minimization over local descriptors from different categories with a shared regularization penalty, which can be formulated as a linear programming problem. The transferred descriptors are further quantized and aggregated to the visual vocabulary. Experiments are performed on PASCAL VOC 2007 benchmark and the quantitative comparisons with several state-of-the-art approaches demonstrate the effectiveness of our proposed approach.

Categories and Subject Descriptors

I.4.7 [**Image Processing and Computer Vision**]: Feature Measurement—*feature representation*; H.3.1 [**Information Storage and Retrieval**]: Content Analysis and Indexing—*indexing method*

General Terms

Algorithms, Experimentation, Performance

Keywords

Local descriptor, semantic information, visual categorization

1. INTRODUCTION

Recently, Bag-of-Features (BoF) model becomes a popular representing scheme for visual categorization [16, 5]. Standard BoF pipeline contains three steps: (1) local descriptors (e.g. SIFT [9] and HoG [6]) are extracted from the images; (2) parameters of the vector quantization algorithm are learned from the training descriptors and each image is mapped from continuous high-dimensional local descriptor space into the discrete histogram of vocabulary with visual words; (3) the histograms are matched based on some metrics, e.g., $L1$-norm or $L2$-norm. In this paper, we focus on the second step of the pipeline.

Standard BoF frameworks use k-means clustering as vector quantization method [16, 5]. k-means and some of its extension methods are trained without class information. Recent works [13, 10] with category labels show the use of semantic information can improve the categorization performance. Specifically, Perronnin et al. [13] construct category-specific histograms in a similar manner to k-means, while Moosmann et al. [10] build a different tree structure for quantization. Here we also aim to develop a category-specific structure for the BoF framework to enhance the visual categorization results.

In this paper, we propose a novel framework for vector quantization by refining the local descriptors. Specifically, we employ a two-step projection on original local descriptors: the category-independent projection for dimensionality reduction, and the category-dependent projection for embedding the semantic information. After projections, we further quantize and aggregate the transferred descriptors to form the visual vocabulary. The highlights of this representation are: (i) the lightweight randomized method for category-independent projection is simple and extremely fast; (ii) the basis for category-independent projection is learned by minimizing the joint loss minimization over local descriptors from different categories with a shared regularization penalty, which can be formulated as a linear programming problem and solved efficiently; (iii) the visual words are generated without running clustering or taking extra training process.

The remainder of this paper is organized as follows. Section 2 reviews the related work on task of vector quantization for visual categorization. Section 3 describes our proposed approach. In Section 4 we demonstrate the quantitative study on PASCAL VOC 2007 dataset. And finally, we conclude our work in Section 5.

2. RELATED WORKS

A number of attempts to improve k-means exist in the literature [12, 14, 13, 10, 17, 11]. As discussed above, we can roughly summarize them into two groups: unsupervised and

†Corresponding author: honglu@fudan.edu.cn
*Area chair: Tat-Seng Chua

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MM'11, November 28–December 1, 2011, Scottsdale, Arizona, USA.
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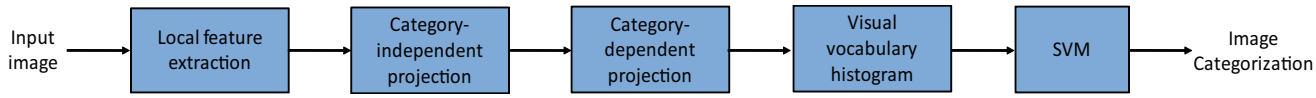

Figure 1: The flowchart of our image representation.

supervised methods. The unsupervised ones try to accelerate the vector quantization speed without semantic labels. For example, Nister et al. [12] replaced k-means by the hierarchical k-means clustering to obtain the tree structure vocabulary. Philbin et al. [14] employed approximated k-means by using the randomized k-d trees for the local feature assignment in the nearest-neighbor search.

Another group of methods incorporate category information into the learning of vector quantization procedure [13, 10, 17, 11]. Perronnin et al. [13] represent images using general histogram and a set of category-specific histograms. Moosmann et al. [10] and Uijlings [17] build discriminative visual word vocabularies using Random Forest methods. Probably most similar to our approach is the work of Ma et al. [11] on combining the idea of Locality Sensitive Hashing (LSH, [1]) and Random Forest, which makes a balance between accuracy and efficacy. Our proposed approach is also simple and effective and can achieve better categorization results.

3. METHODOLOGY

In this section we describe our framework to refine the local descriptors by embedding the semantic information. An overview of our image representation chain is summarized in Figure 1.

3.1 Embedding Semantic Information

Let $\mathbf{X}^c = \{\mathbf{x}_1^c, \mathbf{x}_2^c, ..., \mathbf{x}_{n_c}^c\}$ denote the set of training local descriptors for category c where n_c is the total number of descriptors. The local descriptor $\mathbf{x}_i^c \in \mathbb{R}^d$ is a d dimensional vector extracted from one local region of training images of category c. Our initial motivation is the local descriptor refinement by mapping the descriptors into a more discriminant feature space. Many previous dimensionality reduction or projection-based methods enable the mapping by linear functions. For example, Principal Component Analysis (PCA), one of the standard methods for dimensionality reduction, has been applied to local descriptors [8]. Another potential solution is the randomized projection, which has been applied to image data [3]. Formally, linear projection on the local descriptors can be computed as

$$\mathbf{f}(\mathbf{x}_i) = \mathbf{W}^T \mathbf{x}_i$$

where the $d \times p$ matrix $\mathbf{W}$ denotes the projection parameters (e.g. basis of PCA or randomized basis) and $\mathbf{f}(\mathbf{x}_i) \in \mathbb{R}^p$ is the transformed descriptor. Different with this approach above, our strategy is to get $\mathbf{W}^c$ for each category c separately, i.e.,

$$\mathbf{f}^c(\mathbf{x}_i) = (\mathbf{W}^c)^T \mathbf{x}_i$$

Let $\mathbf{f}^c(\mathbf{x}_i) = (f_1^c(\mathbf{x}_i), f_2^c(\mathbf{x}_i), ..., f_p^c(\mathbf{x}_i))^T$ and we consider the single linear project function $f^c(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$ for late descriptions and $\mathbf{f}^c(\mathbf{x}_i)$ are generated by repeating the procedure p times. In order to construct a more discriminant representation, we aim to learn $f^c(\mathbf{x})$ by a linear classifier determines the relevance of the descriptor $\mathbf{x}$ to category c.

Instead of learning $\mathbf{w}$ directly, a two-step projection is employed here: the category-independent projection for dimensionality reduction, and then the category-dependent projection to embed the semantic information. There are two reasons for incorporating the category-independent projection: (i) category-independent projection reduces the parameter number and makes the framework more efficient; (ii) learning procedure under such methods can embed information about categories. In fact, the two-step projection can be degenerated to single-step one if we set the projection matrix of category-independent projection as an identity matrix.

3.1.1 Category-Independent Projection

We choose random projection for its simplicity and efficiency. Let random functions be $r_j(\mathbf{x}) = \Phi_j^T \mathbf{x} - b_j$ $(j = 1, ..., k)$, where $\Phi_j \in \mathbb{R}^d$ are randomly sampled from normal distribution $\mathcal{N}(0, 1)$, and b_j is the threshold determined based on the information gain. Note that $r_j(\cdot)(j = 1, ..., k)$ is produced in an unsupervised setting and all the categories share the random functions. To find the proper threshold b_j, we group the training descriptors $\mathcal{D}$ with different categories into two subset $\mathcal{D}^+$ and $\mathcal{D}^-$ according to whether $r_j(\mathbf{x}) = \Phi_j^T \mathbf{x} - b_j \geq 0$ or not. We calculate the entropy $E(\mathcal{D}^+)$ and $E(\mathcal{D}^-)$ of the categories in each descriptor subset. The final score is defined as in [15], i.e.

$$Score(b_j) = -\frac{|\mathcal{D}^+|}{|\mathcal{D}|} E(\mathcal{D}^+) - \frac{|\mathcal{D}^-|}{|\mathcal{D}|} E(\mathcal{D}^-) \qquad (1)$$

We randomly generate several candidates for threshold b_j and the one that maximizes Equ. (1) is chosen.

3.1.2 Category-Dependent Projection

Linear project functions $f^c(\mathbf{x})$ are further defined as $f^c(\mathbf{x}) = \sum_{j=1}^k \alpha_j^c r_j(\mathbf{x})$, where α_j^c are the category-dependent parameters, and k is the number of random functions described above.

Intuitively, learning the optimal parameters $\alpha_j^c (j = 1, ..., k)$ can be formatted as follows:

$$\min_{\alpha^c} \sum_{\mathbf{x}_i \in \mathbf{X}^c \cup \mathbf{X}_n^c} L(f^c(\mathbf{x}_i), y_i) + \lambda \sum_{j=1}^k |\alpha_j^c| \qquad (2)$$

where $\mathbf{X}_n^c$ means a descriptor subset randomly selected from training set except for $\mathbf{X}^c$ (note that $|\mathbf{X}^c| = |\mathbf{X}_n^c|$ to balance the weight of positive and negative samples). We set $y_i = 1$ if $\mathbf{x}_i \in \mathbf{X}^c$ and $y_i = -1$ if $\mathbf{x}_i \in \mathbf{X}_n^c$. The first term measures the training errors of each training descriptors (here we use the hinge loss, i.e. $L(f^c(\mathbf{x}_i), y_i) = \max(0, 1 - y_i f^c(\mathbf{x}_i))$). The second term promotes sparsity of parameter α^c.

Inspired by the multi-task feature learning framework [2], we propose to solve Equ. (2) for different categories simultaneously. we consider the following optimization problem

$$\min_{\alpha} \sum_c \sum_{\mathbf{x}_i^c \in \mathbf{X}^c \cup \mathbf{X}_n^c} L(f^c(\mathbf{x}_i^c), y_i^c) + \lambda \sum_{j=1}^k \max_c |\alpha_j^c| \qquad (3)$$

Figure 2: Sample images of the PASCAL VOC 2007 dataset [7].

The left hand term in Equ. (3) is the empirical error across different categories while the right hand term is a shared regularization penalty which suppresses the importance of some random function $r_j(\cdot)$ for all the categories.

The optimization problem in Equ. (3) can be formulated as a linear programming problem:

$$\min_{\alpha,\beta,\xi} \sum_c \sum_{\mathbf{x}_i^c \in \mathbf{X}^c \cup \mathbf{X}_n^c} \xi_i^c + \lambda \sum_{j=1}^{k} \beta_j \tag{4}$$

$$\text{s.t.} \quad y_i^c \sum_{j=1}^{k} \alpha_j^c r_j(\mathbf{x}_i^c) \geq 1 - \xi_i^c, \ \forall c \forall i$$

$$\xi_i^c \geq 0, \ \forall c \forall i$$

$$\beta_j \geq \alpha_j^c \geq -\beta_j, \ \forall c \forall j$$

$$\beta_j \geq 0, \ \forall j$$

3.2 Computing Visual Vocabulary Histogram

Given an image $\mathcal{I} = \{\mathbf{x}_1, ..., \mathbf{x}_m\}$, the visual vocabulary histogram for category c is generated based on category-specified projected descriptors $\{\mathbf{f}^c(\mathbf{x}_1), ..., \mathbf{f}^c(\mathbf{x}_m)\}$ in Section 3.1. Due to design of category-dependent projection, we can employ a function $I : \mathbb{R} \in \{0, 1\}$, i.e.

$$I(f^c(\mathbf{x})) = \begin{cases} 1, & f^c(\mathbf{x}) \geq 0 \\ 0, & f^c(\mathbf{x}) < 0 \end{cases} \tag{5}$$

Since $\mathbf{f}^c(\mathbf{x})$ is p-dimensional, the vocabulary with at most 2^p unique visual words is generated. We then compute the representation for the image $\mathcal{I}$ by counting the number of descriptors assigned to each visual words.

4. EVALUATION

We evaluate the proposed approach on PASCAL Visual Object Classes Challenge (VOC) 2007 dataset [7]. There are 9,963 images in total, including 2,501 images for training, 2,510 for validation, and 4,952 images for testing. The dataset contains annotation of 20 different object categories (as illustrated in Figure 2).

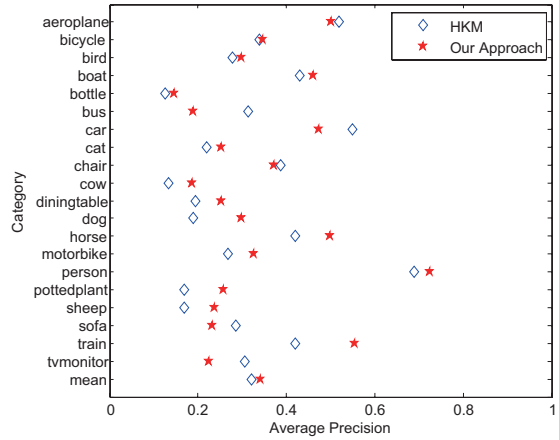

Figure 3: Performance comparison between proposed method and unsupervised method.

4.1 Implementation

For each image in the dataset, we use dense sampling strategy to find the local regions. Then SIFT descriptor is applied to generate the 128-dimensional local feature. About 2,500 local descriptors are extracted for each image. We empirically set the number of random functions for each $f^c(\mathbf{x})$ as $k = 32$ and the dimension of $\mathbf{f}^c(\mathbf{x})$ is $p = 12$. The visual word filtering strategy [11] is applied and finally we have a vocabulary with $1,000$ selected visual words.

To get the object categorization accuracy, support vector machine (SVM) with χ^2 kernel is used in our experiment: $d_{\chi^2}(f_1, f_2) = \frac{1}{2}\sum_{i=1}^{d} \frac{(f_{1i}-f_{2i})^2}{f_{1i}+f_{2i}}$, where f_1 and f_2 are two d-dimensional image features. Specifically, libSVM [4] is used to train and test the categorization models.

The categorization performance is evaluated using average precision (AP), defined as $AP(c) = \frac{1}{m_c}\sum_{k=1}^{n} p(k) I_c(k)$, where n is the dataset size, m_c is the image number of category c, $p(k)$ indicates the precision at position k, $I_c(k) = 1$ indicates the the image with rank k-th belonging to category c and $I_c(k) = 0$ otherwise.

4.2 Results and Discussions

Our approach is compared with several state-of-art methods, containing both unsupervised and supervised strategies.

Unsupervised method We employ the hierarchical k-means (HKM, [12]) on SIFT descriptors. 200,000 local descriptors are randomly chosen from the training images. In our experiments, the hierarchical k-means builds a vocabulary tree with the branch factor (number of children of each node) $k = 10$ and maximum number of levels $L = 3$; then 1,000 clusters are obtained for the comparison. Comparison between our proposed method and hierarchical k-means is shown in Figure 3. We can clearly see the superiority of proposed method over hierarchical k-means with the same number of vocabulary. The improvement of mean average precision is 2%. Our method performs better for 14 out of the 20 categories, which suggests the effectiveness by embedding semantic information.

Supervised methods We also follow the setup of Random Forest (RF, [17]) and Randomized Locality Sensitive

Vocabularies (RLSV, [11]) to construct the visual word vocabulary. The same 200,000 local descriptors as that for hierarchical k-means are used for training the trees. Here each vector also has a category information from the annotation of its original image. For RF approach, learning the supervised tree is done recursively and finally each tree is with maximum number of levels $L = 3$ and about 1,000 visual words. Figure 4 illustrates the categorization results between RLSV, RF, and our approach for all the categories. Our method performs better than RLSV for 14 categories, and RF for 13 categories out of 20 categories. Overall, the results demonstrate that our method outperforms previous supervised methods.

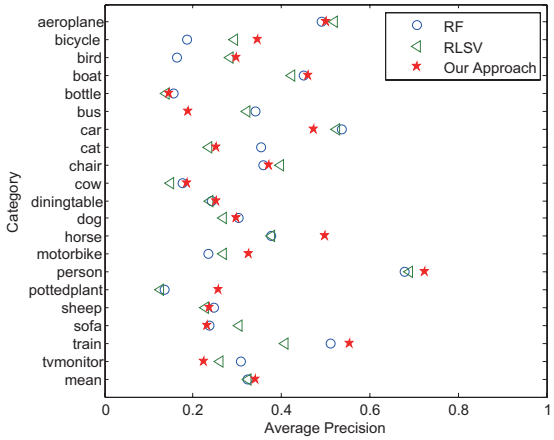

Figure 4: Performance comparison between proposed method and supervised methods.

5. CONCLUSION

In this paper, we have introduced a method to embed semantic information into local descriptors for visual categorization. The local descriptors are integrated by a sequence of category-independent and category-dependent basis. The category-independent basis are selected from random projection, and the category-dependent basis is learned by an optimization problem which minimizes the joint loss minimization over the descriptors from different categories. The transferred descriptors are further quantized and aggregated to the visual vocabulary for categorization. Preliminary evaluation on PASCAL VOC 2007 dataset have confirmed the effectiveness of our method.

Currently, we only implemented a basic system with only 20 categories and the linear programming problem Equ. (4) can be solved efficiently. However, since the number of variables and constraint conditions of the problem are proportional to the category numbers, solving it may be computationally expensive with the increasing of categories. Our future work will incorporate the hierarchy taxonomy structure on large number of categories. Another potential direction may be the using of the non-linear function (instead of the random linear functions in Section 3.1.1), which may be beneficial to some object categories (as suggested in [18]).

Acknowledgment

This work was supported in part by 973 Program (Project No. 2010CB327900), National High Technology Research and Development Program of China (No. 2009AA01A346), Natural Science Foundation of China (No. 60875003 and 60873178), 211-Project Sponsorship for Young Professors. Y. Zheng is also supported by the outstanding doctoral research funding of Fudan University.

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